Lectures on Relativity

Non-Inertial Special Relativity and General Relativity

Licence 3 - Master - Engineering schools

2025 - Mathieu ROUAUD

Foreword

This document compiles the courses taught over one semester at *Universitad Mayor San Simon* in Cochabamba, Bolivia. There are many excellent books and PDFs on general relativity, but here is what characterizes and can sets this one apart:

1. Numerous exercises with detailed answers accompany the lectures.

2. Special relativity is often studied only in the inertial context, but in the absence of gravity, it also applies to non-inertial reference frames. We study the cases of uniformly accelerated rectilinear motion and rotating frames. This is a very rich subject that allows for a smooth transition to general relativity, by familiarizing oneself with the concept of coordinate time, tensor calculus, and Lagrangian formalism.

3. In cartography, we use various metrics to represent the sphere on a plane. The shortest path is no longer necessarily a straight line. We use this analogy to understand geodesics that maximize proper time in general relativity.

4. Some chapters are based on a research paper and help to prepare to adopt the researcher's approach.

Good reading!

Always happy to receive mail from my readers: ecrire@incertitudes.fr

For special relativity, reference will regularly be made to the book by the same author:

[SR] *Special Relativity, A Geometric Approach, followed by the conference Interstellar travel and antimatter., 2020, 536 pages.*

Files on github.com/Mathieu-ROUAUD

First Lecture

Special Relativity

Mathieu : Hello. Let's begin.
Since birth, we have lived in the social fiction of absolute time. However, as we will explain today, time is relative. This new concept needs to be clearly explained, otherwise it can lead to a lot of confusion. For example, the relativity of time in no way calls causality into question. It is impossible to travel into the past, or for someone from the future to visit us. However, as we will see later, we can travel into the future of others.

Arkaitz : Sir, but why not start directly with general relativity? We have already had special relativity classes since the bachelor’s degree.

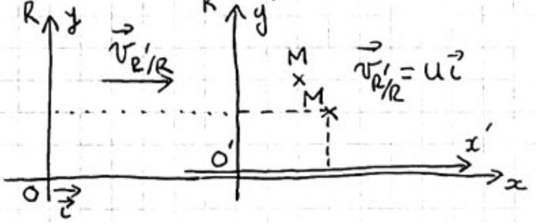
Mathieu : Indeed, it may seem unnecessary to repeat what we already know. But every teacher has a different approach to special relativity. Today, for example, we will look at a visual tool for determining time dilation that you have never seen before: the time triangle. In addition, this will be an opportunity for us to introduce notation, and in the third lecture we will introduce the tensor approach necessary for general relativity. With a new definition of special relativity that breaks away from the historical bias of light, and which, thanks to symmetries, has made it possible to construct the Standard Model.

Cristian : All right, and when will we begin general relativity itself?

Mathieu : We have a total of 18 lectures, only the last 6 of which will deal exclusively with general relativity. Before that, we need to complete your lectures on special relativity. You are not yet familiar with tensor calculus, and, for example, we are going to put Maxwell's equations into tensor form, which will then guide us by analogy for gravitational waves. We will discuss special relativity in accelerated reference frames, which will also be new to you. This will be an opportunity to introduce metric vector spaces and the Lagrangian.

Any other questions?

Okay, then... Historically, the concept of relative time is the result of a contradiction between two theories. On the one hand, we have Newton's theory from 1687, which describes the motion of objects and matter in general, and on the other, Maxwell's theory from 1864, which describes the behavior of light in terms of electromagnetic waves. These two theories, which have been thoroughly verified experimentally, state their laws in inertial reference frames. However, the first predicts the additivity of velocities, which contradicts the second theory's assertion that the speed of light in a vacuum is constant.



at t=0, O=O'

Newton :

Galilean Transf. $\begin{cases} x' = x - ut \\ y' = y \\ z' = z \end{cases}$ Additivity of velocities $\begin{cases} v_{x'} = v_x - u \\ v_{y'} = v_y \\ v_{z'} = v_z \end{cases}$

Maxwell (vacuum):

• Non invariance of eq. under Galileo

$R: \vec{\nabla}.\vec{E} = 0 \xrightarrow[\text{SR p433}]{\text{dem.}} R': \vec{\nabla}'.\vec{E}' \neq 0$

• $\vec{\nabla} \wedge \vec{E} = -\partial \vec{B}/\partial t$ and $\vec{\nabla} \wedge \vec{B} = \mu_0 \varepsilon_0 \, \partial \vec{E}/\partial t \Rightarrow \Box \vec{E} = \vec{0}$

$\frac{1}{c^2}\frac{\partial^2 \varphi}{\partial t^2} - \Delta \varphi = 0$ with $c = 1/\sqrt{\mu_0 \varepsilon_0}$

An idea, to put everyone in agreement and save the two theories, would be that light propagates in a particular referential named *ether*. However, the experiments of Michelson-Morley, carried out as early as 1887, prove that ether, supposed medium of propagation of light, does not exist.

Which leads Einstein to propose a new theory in 1905. This theory is much more satisfactory because it integrates matter and light in the same framework. Here are the two main **postulates** of Einstein's theory of special relativity:

1. *Principle of relativity*: The laws of physics are the same in all inertial frames.

    Note the word *physics* and not *mechanics* as with Newton:
    physics = mechanics + electromagnetism = matter + light

2. *Universality of the speed of light*: The speed of light in a vacuum is the same for all inertial observers.

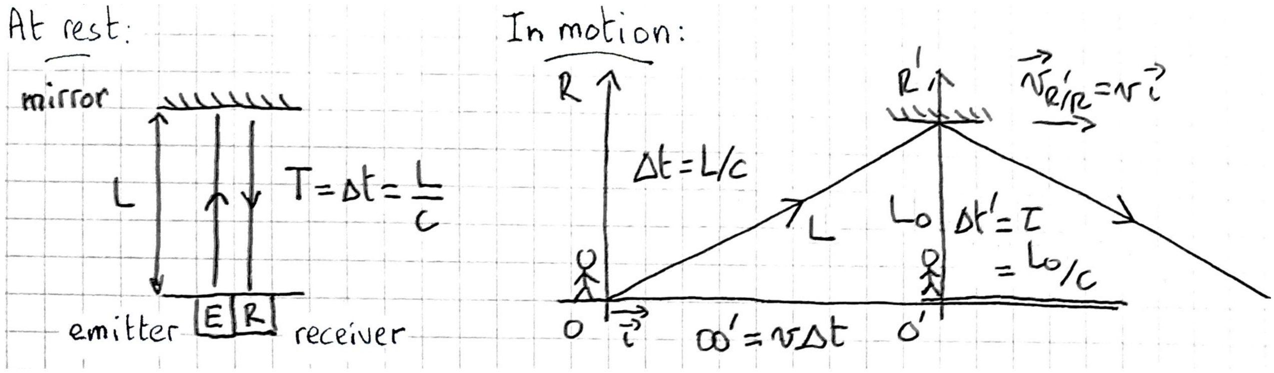
    c = 299 792 458 m/s (universal constant fixed by special relativity)

Arkaitz : It's surprising that light plays such a central role. Gravitational waves, as verified in 2017, also go to velocity c, and any particle of zero mass. I find it strange that the content and the container are linked.

Mathieu : Yes, the second postulate is actually useless. It is a consequence of simple assumptions about symmetries: 3 dimensions of space, one of time, homogeneity of time and space, isotropy of space, principles of relativity and causality. These assumptions seem natural and allow to find the Lorentz transform in a few pages of calculation[1]. The constant c then appears as a structural constant specific to space-time, unrelated to any physical object.
It is interesting to note that Newton had discovered the mathematical tools of differential calculus that could have allowed him to find this most general change of coordinates between two inertial frames of reference. Without realizing it, he restricted himself to the limit case where c is infinite, and space and time, absolute and disconnected.

Let's continue nevertheless with the 2 historical postulates of Einstein and describe the ***light clock***. Let's imagine a clock that uses light rays. A light ray is emitted, reflects on a mirror and returns to the same point after a *2T* duration:

1 Well explained and demonstrated by Jean-Marc Lévy-Leblond, « One more derivation of the Lorentz transformation », American Journal of Physics 44 (3) 271 - 277 (1976).



The clock is at rest in *R'* and moves at the velocity $\vec{v}$ with respect to *R*. The distance traveled by the ray in *R* is therefore larger, while keeping the speed constant:

$$\text{so}\quad L^2=L_0{}^2+OO'^2 \quad\Rightarrow\quad c^2\Delta t^2=c^2\Delta t'^2+v^2\Delta t^2 \quad\Rightarrow\quad \Delta t=\frac{1}{\sqrt{1-v^2/c^2}}\Delta t'$$

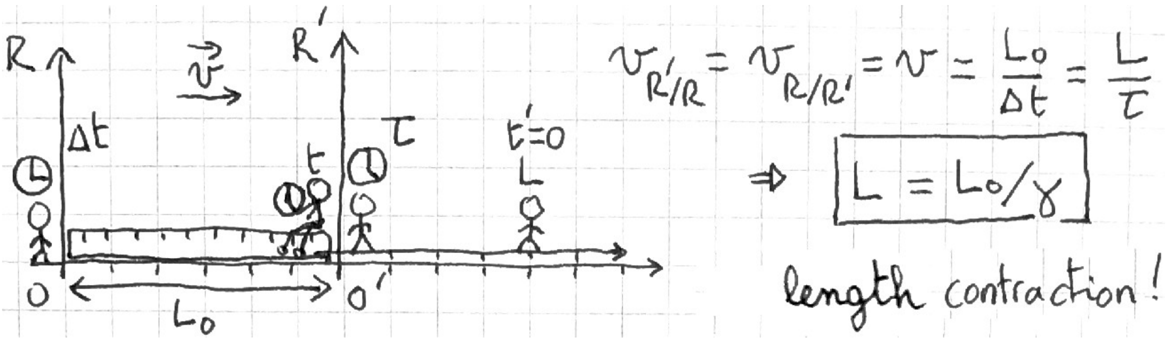
$$\gamma=\frac{1}{\sqrt{1-\beta^2}} \quad , \quad \beta=\frac{v}{c} \quad , \quad 0\le\beta<1 \quad , \quad \gamma\ge1 \quad \text{then} \quad \boxed{\Delta t=\gamma\tau} \;\Rightarrow\; \textbf{time dilation}!$$

Adriana : ... so an observer of *R* sees the clock of *R'* going more slowly, and it's relative, that of *R'* also sees a clock at rest in *R* going more slowly.

Mathieu : Yes, the situation is symmetrical. But, no, it's not what is seen but what is measured. We will see later the methods for measuring times and distances. Here the time dilation is the same for frames that move closer or further away. On the other hand, for what we see, it is different, because it is necessary to take into account, in addition, the propagation of light. We will see that in a next class, it's the Doppler effect.

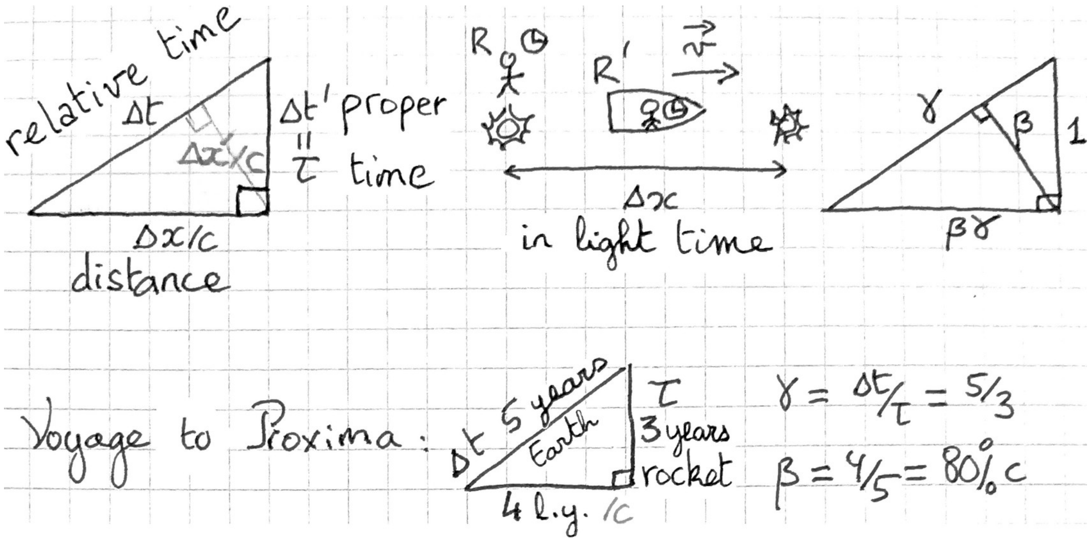
Time dilation can make one think of a spatio-temporal perspective effect. An analogy, from where I am, I can "hold" a student at the back of the room between my thumb and forefinger, he is very small, like a Smurf! But, he also sees me as tiny, it's mutual. In this case, we have a spatial perspective effect. And, of course, we actually both have the same size, it's just an illusion.
In the case of relativity it is more complex, because, unlike space, time is not isotropic.

Where were we? Ah yes! Moreover, time dilation is associated with a **length contraction**. Let's consider a ruler at rest in *R*, and, as before, a clock at rest in *R'*:



To measure a length in a given reference frame, it is necessary to identify the position of each end at the same moments.

Relying on the Einstein's light clock, let us now present the **time triangle** that summarizes these effects:



We are ready to address the twin paradox: imagine twins who are 20 years old, one stays on Earth, and the other goes in his spaceship at a speed of 80% of *c* for Proxima Centauri located 4 ly. The traveling twin arrives, on the exoplanet *Proxima b*, 3 years later, while it has been on Earth for 5 years. When he returns to Earth, it flows again 3 years in the ship and 5 years on Earth. So, when they meet again, the traveling twin will be 26, while the sedentary twin will be 30! If we imagine the situation symmetrically, it seems paradoxical. But one is undergoing acceleration to return, unlike the other. And ultimately, this is not just an illusion, a perspective effect, but a physical and measurable difference.

Luna : Here, the rocket always travels at the same speed, which is not realistic! You can't go from 0 to 80% of *c* instantly, and even less turn around suddenly by reversing speed. However, it is because of these accelerated phases that the traveler ages less than someone on Earth. Because the rest of the time, the situation is symmetrical. Do you think it is conceivable that such a rocket could be built one day?

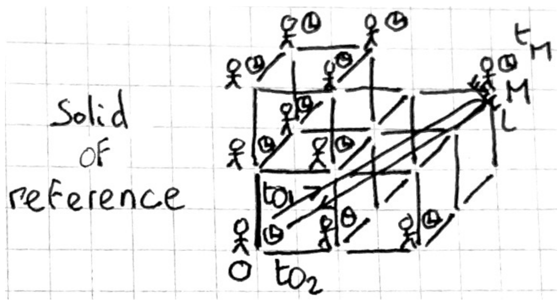
Mathieu : Yes, excellent points. For greater realism, in the next lesson we will consider a uniformly accelerated rocket. The rocket's frame of reference is then non-inertial, and Einstein's postulates do not apply. The metric is no longer Minkowskian but of Rindler. Such a rocket is conceivable, but with many “ifs”. If we were able to collect, store, and use antimatter efficiently, a rocket the size of a *Saturn V* would enable a trip to *Proxima b* with an acceleration of 1 *g*, taking 3 years for the traveler and 5 Earth years.

To conclude today, let's look at a fundamental point: the definition of an inertial reference frame in special relativity, the only reference frames in which the postulates are verified.

To locate ourselves in space, we proceed as in Newtonian mechanics. We place one-meter rulers perpendicular to each other to form a grid that covers the entire space. We also want to have clocks at each node of the grid, with all clocks constantly synchronized. In addition, we imagine an observer at each node that records events occurring at its level using labels (ct, x, y, z).
The difference with classical mechanics concerns the synchronization of clocks. If you place all the clocks at node *O*, set them to zero, and then move them to their respective nodes, they will become desynchronized during the motion, because they will then have a non-zero velocity and their time will dilate.

Hence ***Einstein's synchronization method***: the clocks are first placed at their nodes. Arbitrarily, the clock at O, H_O, is chosen as a reference. At time t_{O1} on H_O, the observer at O emits a light beam toward a clock at M to be synchronized. The ray is reflected by a mirror at M and returns to O. When the ray is reflected, the observer at M notes the time t_M indicated by his clock. When the ray returns, O notes t_{O2}. O communicates to M the values t_{O1} and t_{O2}, and M deduces the synchronized t_M.

By symmetry of the trajectories between the outward and return paths (isotropy of space):



$(t_M)_{sync}=\dfrac{t_{O1}+t_{O2}}{2}$, H_M to be advanced by $(t_M)_{sync}-t_M$.

Cristian : Okay, so the clocks are initially synchronized. But what guarantees that they will remain so?

Mathieu : The homogeneity of space and time required in an inertial reference frame. No point should be favored; due to symmetry, there is no reason why one clock should start behaving differently from the others.
A major advantage of special relativity is that we have an experimental method for determining whether a reference frame is inertial: once synchronized, clocks remain synchronized. If the reference frame is non-inertial, they become desynchronized over time.
In Newtonian mechanics, time is absolute, and a reference frame is Galilean (inertial) if Newton's laws are verified, but the laws are only verified in a Galilean reference frame, so the definition is circular. In classical mechanics, if an object has a curved trajectory, we do not know whether this is due to the action of a force or the non-inertial nature of the reference frame.
In relativity, we use the clock crystal, and this clarifies the issue.

When we talk about measuring time and length, we are referring to this reference solid and its associated clock crystal. Each reference frame has a set of observers, each with their own rulers and clocks. Following an experiment, the observers gather all the data on the various events they have collected to reconstruct what happened[2].

Consider a ruler of length L_0 and a clock with period τ, both at rest in R. When a frame R' moves with the velocity $\vec{v}$, an observer in R' will sometimes coincide with a given observer in R. For example, there are two observers A_1 and A_2 of R at rest at each end of the ruler, and in R' at the same time t' two observers A'_1 and A'_2 will coincide with them:

$$\text{for all } t,\ L_0=A_1A_2 \quad ; \quad \text{same } t',\ L=A'_1A'_2=\sqrt{(x'_{A'_2}-x'_{A'_1})^2+(y'_{A'_2}-y'_{A'_1})^2+(z'_{A'_2}-z'_{A'_1})^2}.$$

For the clock, an observer of R is constantly at rest, and two observers of R' will coincide, one at the beginning and the other at the end of the period:

$$\text{same } x,\ \tau=t_2-t_1 \text{ (one clock)} \quad ; \quad \text{different } x',\ \Delta t'=t'_{A'_2}-t'_{A'_1} \text{ (two clocks).}$$

And it is according to these protocols that we have $L=L_0/\gamma$ and $\Delta t'=\gamma\,\tau$.

Thank you for your attention.

2 It should also be noted that clocks at rest are thus synchronized in a given inertial reference frame, but *a priori* they do not appear to be synchronized with respect to another reference frame in which they are in motion.

Summary of the notions covered today:

Lecture 1 - SPECIAL RELATIVITY

- Einstein's Postulates
- Light clock: Time Dilation
- Length Contraction
- Time Triangle
- Twin Paradox
- Definition of an inertial frame: Einstein's synchronization method

To review and practice:

- This lecture covers Chapter 1 of the SR book, which includes exercises 1 to 8 with their answers.

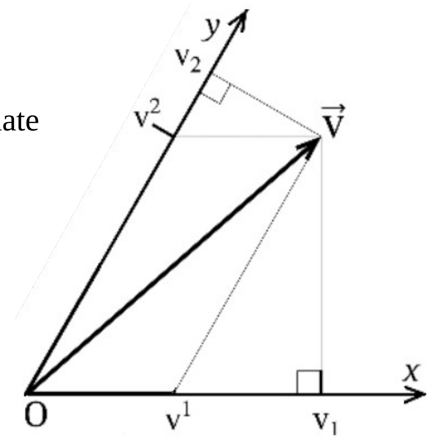
Second Lecture

Minkowsky diagrams

Mathieu : Hello.
Today we present a graphical tool proposed in 1908 by Hermann Minkowski.
In relativity, position in space-time is specified by four coordinates grouped together in the position four-vector $\tilde{x}$ with components x^μ where $\mu=0,1,2,3$. Here, the index is placed above, and it is said to be *contravariant* (it is not a power!).
In this course we will take: $\tilde{x} = (x^0=ct\,,\ \ x^1=x\,,\ \ x^2=y\,,\ \ x^3=z)$.
When a particle moves through space-time, the set of successive events constitutes a *worldline* in a 4-dimensional space of a reference frame *R*.

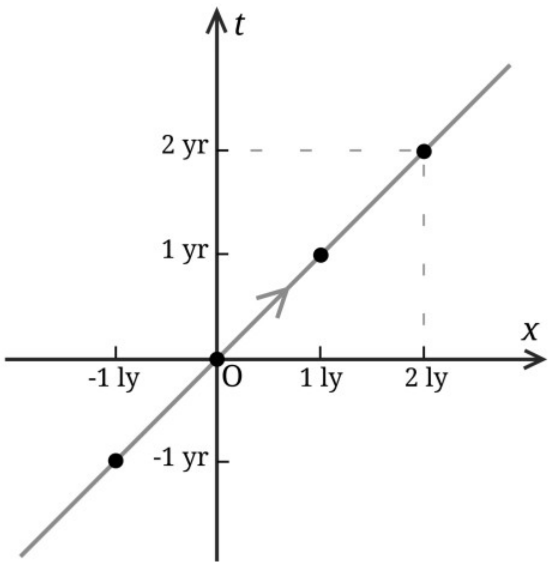
Adriana : ... what if we put the index at the bottom?
Would that change anything?

Mathieu : For a vector of $\mathbb{R}^3$ projected on an orthogonal coordinate system, this does not change anything, but in general it is different.
In a non-orthogonal coordinate system, there are two possibilities for projecting a vector: the contravariant components v^μ are projected parallel to the axes, and the *covariant* components v_μ are projected perpendicular to them:



More detailed explanations will be given in the next lesson on vector spaces.

So, an event is denoted *E(ct, x, y ,z)*, and in many cases it is possible to limit ourselves to a straight-line motion and a single spatial dimension: *E(ct, x)*.
In a Minkowski diagram, the time axis *t*, or *ct*, is always vertical. On the *x*-axis, the units of distance are light-times. As *c* is a universal constant, it can be used to unambiguously match a time to a length. For interstellar travel, we choose the year and the light-year.
An object at rest at *O* is represented by a vertical line, the time axis, an worldline pointing upwards, not only the event *O(0, 0)* at the origin for *t=0.*

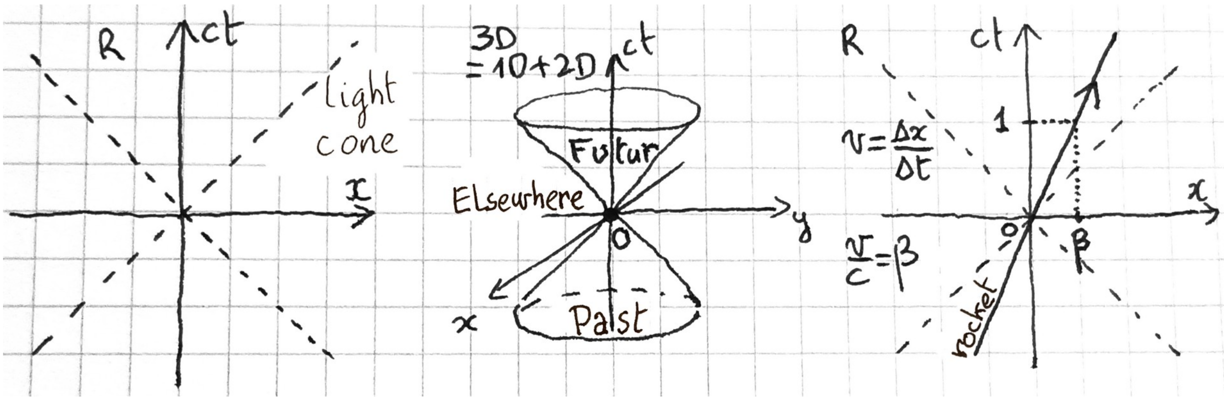


A light pulse traveling to the right and at *O* at *t=0*, will be at one l.y. in one year, and was at *x=-1 ly* one year ago, from which the light's worldline represented in gray according to the first bisector.

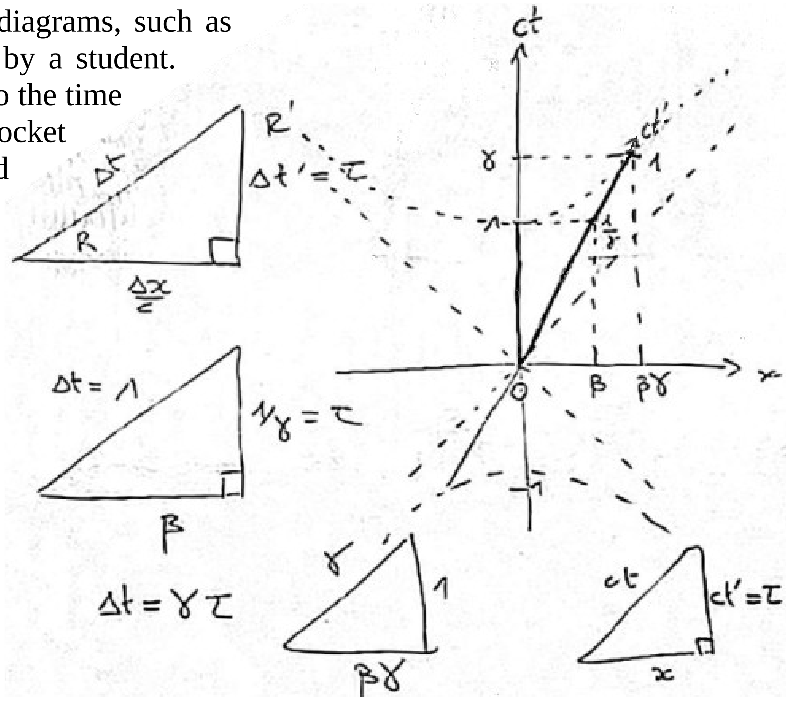
On a space-time diagram, the worldlines of light rays passing through *O* are always represented by dotted lines. They define the light cone (next page).

Arkaitz : Actually, the cone only appears in 3D!

Mathieu : Yes! And, in general, it's 4D with a hypercone!



On the drawing on the right, you see the worldline of an object moving at constant speed *v* along the *x*-axis. For *ct=1*, you can see its speed β, as a percentage of *c*. From *O*, all points inside the cone can be connected by a worldline, but there is no possible connection with those outside because the speed is less than *c*.



The time triangle can also be linked to diagrams, such as this picture of a board taken in Bolivia by a student. The rocket's worldline in *R* corresponds to the time axis *t'=τ* in the reference frame *R'* of the rocket at rest. And according to the second triangle, we read, in *R*, *t'=1/γ* when *t=1*. And in the third triangle, multiplied by a factor γ, time dilation appears explicitly. We have two segments of length 1, which appear to be of different lengths, along *ct* or *ct'* !
The reason is that we are not in a Euclidean space but in a hyperbolic space. Indeed, according to the last triangle:

$$c^2t^2 - x^2 = c^2\tau^2$$

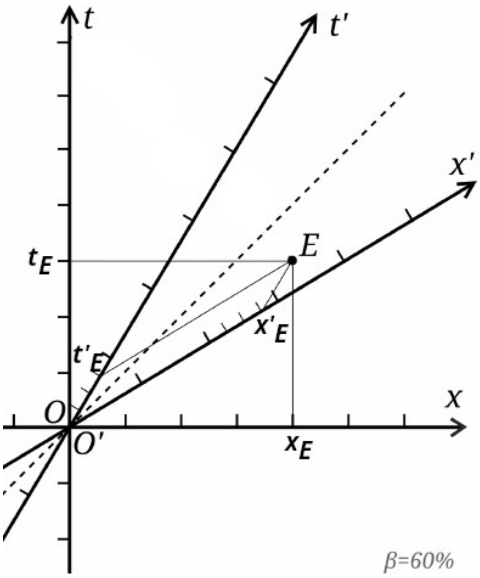
We recognize the equation of a hyperbola. Here, the invariant is the proper time τ, and the curves of constant *proper time* are hyperbolas. In Euclidean geometry, length is invariant, and we are at a constant distance on a circle: $x^2 + y^2 = L^2$. Representing Euclidean geometry on a sheet that is itself Euclidean is fairly simple, but hyperbolic geometry on a Euclidean sheet requires careful attention.

Adriana : But... that's very strange! The lengths are not what we think they are. And the angles?

Mathieu : Also... For example, for any point *M* on the hyperbola, the line *(OM)* is locally orthogonal to the hyperbola, and not only when *M* is on the *ct* axis. By analogy in Euclidean geometry, the radius of a circle is always perpendicular to the circle. We will see later how to perform a scalar product in Minkowski space.

Let us now represent the twins' experience in a diagram (next page). Our reasoning is based on the galactic reference frame, where the stars are assumed to be fixed. The worldline of our sun, and therefore of the twin on Earth, corresponds to the time axis. That of Proxima Centauri is parallel to it, such that *x=4 ly*. For the twin traveler's round trip, his broken world line clearly shows the evolution of his proper time; he is 4 years younger than his brother when he returns.

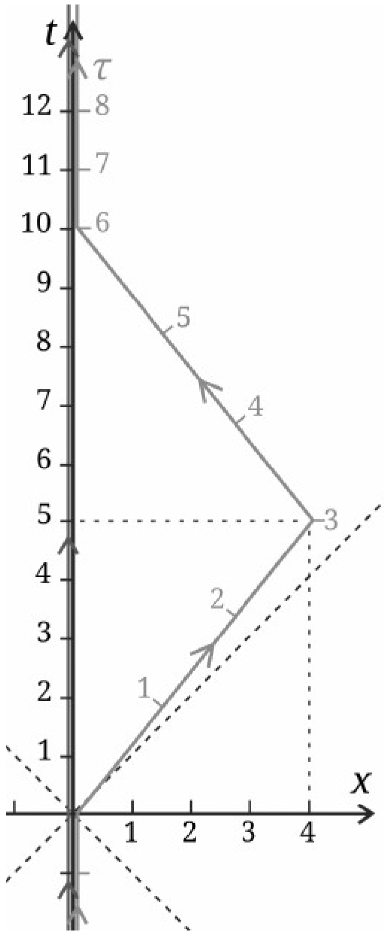
Let's see how to perform a ***change of reference frame***. For a frame R' moving at a velocity $\vec{v} = v\,\vec{i}$ with respect to R, the axis ct' corresponds to the world line of O'. For the axis x', we use the invariance of the speed of light: the axes x' and ct' must be symmetrical with respect to the light cone.



$E=(ct_E, x_E)=(ct'_E, x'_E)$

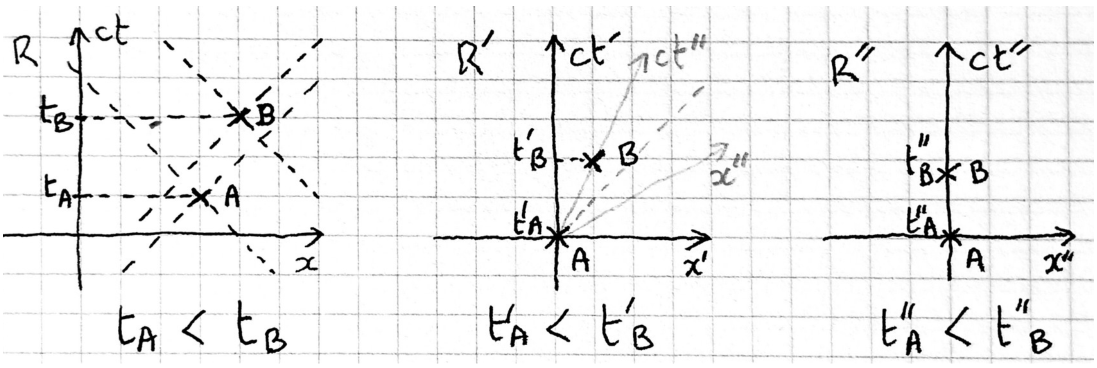
$(3\ ;\ 4)_R=(0{,}75\ ;\ 2{,}75)_{R'}$

The frame R is orthonormal. An inexperienced Euclidean mind might think that R' is not orthogonal. However, it is: two lines that are symmetrical with respect to the cones of light are orthogonal.
Furthermore, if we visualize the hyperbolas, we understand that the norm is the same.



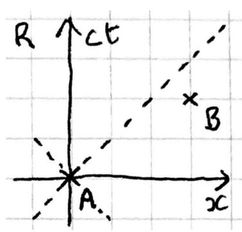
Now let's talk about ***causality***:
If event A precedes event B in R, is this true in all reference frames R'?

<u>First case</u>, A and B are in their respective light cones:



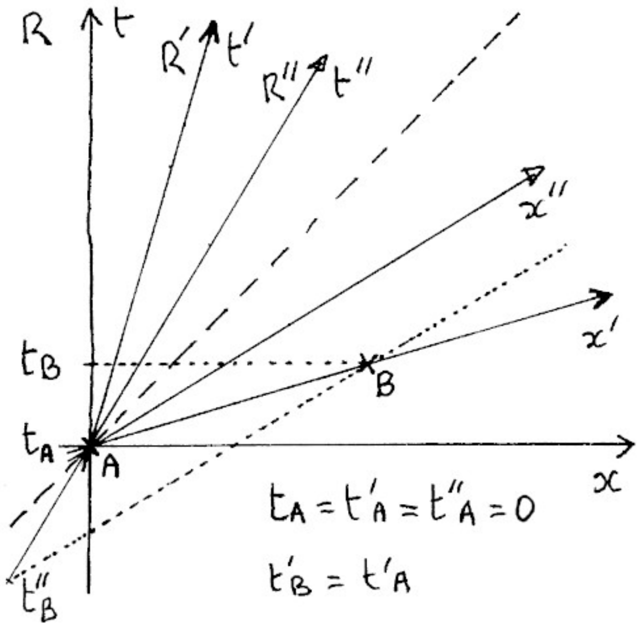
From R to R', it is a simple translation. Then, we can always perform a *Lorentz transformation* from R' to R'', such as A and B are at rest. The chronology is respected, and we can return to a frame R at any velocity relative to R''. Also in R'', this is the simplest case for measuring dates: a single clock is needed, and the anteriority of A over B is measured directly. It follows that: $\Delta t''_{AB} = \tau < \Delta t'_{AB} = \Delta t_{AB}$, where the proper time τ is the minimum duration.

<u>Second case</u>, A and B are not in their cones:
In R, A comes before B because $t_B > t_A$.
In this case, it is impossible to find a frame of reference in which A and B are at rest, because there is no worldline connecting the two events.
In R', such that $x'=(AB)$, see next page, A and B are simultaneous!
In R'', A comes after B!
But whatever the frame of reference, B is outside the cone of A, and has no causal link with it.



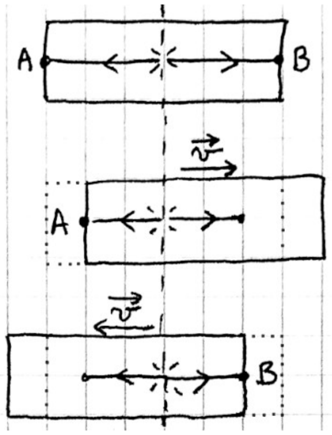
Conclusion, when two events have a possible cause-and-effect relationship, the chronology is respected for all observers, and the principle of causality is therefore verified.
When two events cannot influence each other, it does not matter whether one occurs before the other or whether they occur simultaneously. The chronology then depends on the observer and has no consequence on real life. They have no causal link, so the principle of causality (preservation of the direction of the causal link) does not need to be verified.

Thus, in relativity, causality is just as rigorously verified as in Newtonian mechanics. And the fact that time and simultaneity are relative does not change anything.



Luna : I remember the example where a flash of light is emitted in the middle of a train car, and the light rays reach both ends of the car at the same time. But if the car moves to the right, since the speed of light is constant, the left ray arrives first. And if it moves to the left, it's the right ray. The order of events changes!

Mathieu : Yes, good example. Whereas in classical mechanics simultaneity would be verified in all reference frames, due to the additivity of velocities, one ray would travel at *c*-*v* and the other at *c*+*v* compensating for the motion of the wagon.

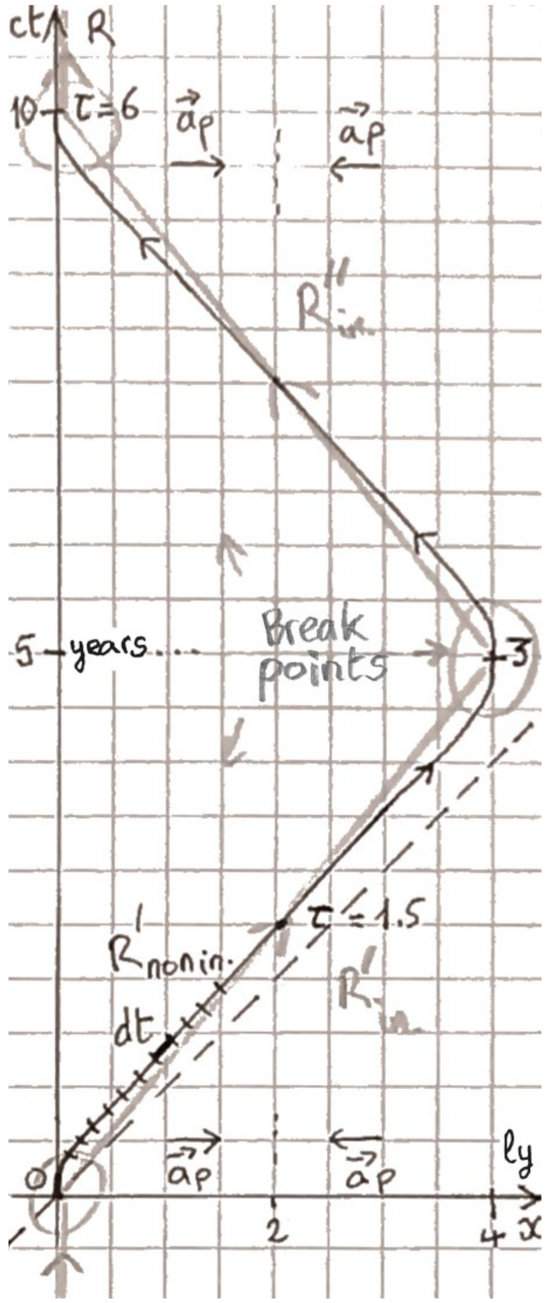


To conclude today, let's return to the twin paradox.
For greater realism, we replace the broken line with non-physical angular points, with a curve with smooth variations (next page). The rocket travels along a straight line at a constant proper acceleration $\vec{a}_p$, and turns around halfway to arrive at Proxima at zero speed. The passengers experience an artificial gravity $\vec{g}=-\vec{a}_p$. The rocket's reference frame *R*' is non-inertial, nevertheless at any moment we can make a second inertial rocket with the same velocity $\vec{v}$ and zero acceleration coincide. For a short duration, as long as the rockets are roughly immobile relative to each other, the clocks of the two rockets will keep the same pace. These ideas that make it possible to deal with non-inertial frames of reference in special relativity are summarized in the following two new postulates:

3. *Equivalence principle*: Locally, at any point in space-time, there is always a coincident inertial reference frame.

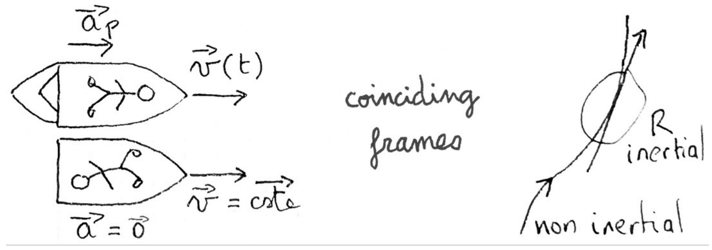
This frame is locally minkowskian.

4. *Clock hypothesis*: Two clocks that coincide with the same speed will have the same time dilation, regardless of their accelerations or rotations.



The time elapsed in the rocket can be calculated from the frame *R*. On a worldline:

$$dt = \gamma\, d\tau \qquad \text{and} \qquad \tau = \int d\tau = \int \sqrt{1 - \frac{v^2}{c^2}}\, dt$$

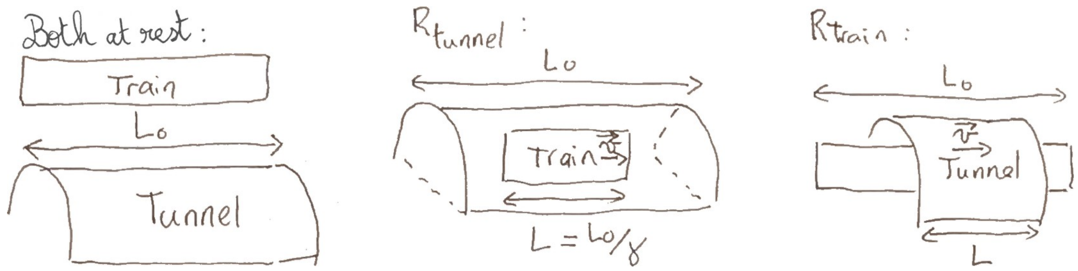


Cristian : Will these new postulates remain true in general relativity?

Mathieu : Yes. And for gravitation, we will complete these four postulates by adding *Einstein's equation*, which links the *metric* to the *energy-momentum tensor*.

The formula for *τ* shows us how, from the perspective of the observer on Earth, the traveler ages more slowly than he does. *dτ* is smaller in relation to *dt* as *v* increases. On the worldline of accelerated motion, it is mainly outside the breakpoints zones that the slowing of aging relative to Earth is most significant.

Ah, but since we still have a few minutes left, we can finish with a nice paradox, involving a relativistic train passing through a tunnel. Both the train and the tunnel have a rest length of $L_0 = 1000$ m. Let's assume that the tunnel is at rest and that the train is arriving with a γ of 2. The train, contracted by a factor of 2, easily enters the tunnel. We can even close the doors at both ends of the tunnel at the same time to trap the train for a microsecond!
What about from the point of view of the train at rest? Now it is the tunnel's turn to be contracted by a factor of 2, and it is impossible to put the train inside...
How can we resolve this apparent contradiction?



Summary of the notions covered today:

Lecture 2 - MINKOWSKI DIAGRAMS

- Event and worldline
- Light cone and hyperbolic space
- Change of reference frame
- Principle of Causality
- Relativity of Simultaneity
- Back to the Twin Paradox
- Clock Hypothesis
- Equivalence Principle
- Train Paradox

To review and practice:

- Chapter 2 of the SR book, exercises 1 to 4 with their answers, and Chapter 3.
- A worksheet follows.
- Represent the paradox of the flash of light emitted in the middle of a train car, then that of the train and the tunnel, using Minkowski diagrams.

Exercises

Time dilation and spacetime diagrams

Exercise 1

In the galactic year 2125, you undertake the Sun-Sirius voyage to study this binary stellar system.
After 8 years in your spaceship, you land on an exoplanet around Sirius A.
In what galactic year are we then, and what was the speed of your ship?
In 2128, three years after you left, the earthlings send you a radio message.
When do you receive this message?
When you arrive on the planet, you send a picture of your new home. When will they receive it?
Draw a Minkowski diagram that shows the various worldlines.

Sun-Sirius distance in the galactic frame of reference: 9 ly.

Exercise 2

Carlos, 40, has to travel, for a secret mission, to the star *Kappa Ceti*, 30 ly away in the equatorial constellation of *Cetus*. The Carlos' vessel is a β95 model. When he left, is daughter Maria, is 15 years old. Maria sends a letter each month to her father with an electromagnetic wave. How often does Carlos receive letters during his trip to *Kappa Ceti*?
Just arrived at *Kappa Ceti*, Carlos came back to the Sun. Show the Minkowski diagram.
When they meet again, how old are the daughter and the father?

β95 model: $\beta=3/\sqrt{10}$

Exercise 3

In Japan, the Shinkansen train takes 3 hours to travel from Tokyo to Shin-Aomori, 675 km away. Initially, two atomic clocks are synchronized at Tokyo. One remains on the station, and the other makes a round trip to Shin-Aomori on the train. Back to Tokyo, what is the desynchronization between the two clocks? Is it measurable? The atomic clocks accuracy is 0.5 ns/day.

The terrestrial reference frame is supposed inertial.

Exercise 4

A plane flies around the Earth straight on the equator toward the east, covering 40,000 km at 1000 km/h ground speed. At the beginning, two atomic clocks at rest are synchronized. One remains on the ground, and the other makes the round trip on the plane around the world. Back to the initial point, what is the desynchronization between the two clocks? What if the plane travels toward west?

Altitude variations are ignored. Geocentric frame considered inertial.

Exercise 5

What is the probability for a muon created at 15 km with a vertical velocity of 0.995c to reach the sea level? Chacaltaya, 5240m?

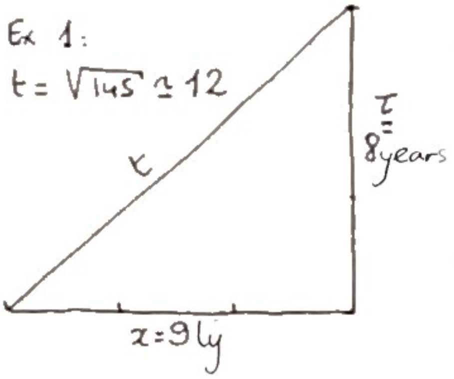
Muon mean life at rest: 2.2 μs

Answers

Lectures 1 & 2

Ex 1:

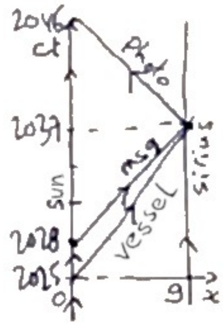
$t = \sqrt{145} \simeq 12$



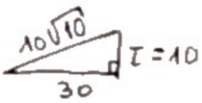
Arrival in 2125+12 = 2137
speed ≃ 9/12 c ≃ 75% of c

Message received the same year than the arrival : 3+9 = 12

Pic received in 2046 (+9)

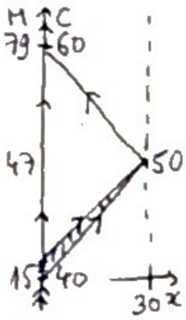


Ex 2:



$\gamma = \sqrt{10} \simeq 3.16$

Back, Maria has 79 and Carlos 60.



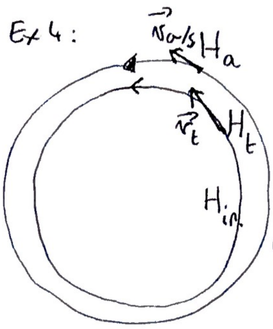
The letters sent between the ages of 15 and 16,6 by Maria will arrive before Carlos' arrival on Sirius : 19 Letters

Ex 3: $\Delta t = \gamma \tau$

$\Delta t - \tau$
$= \Delta t\,(1 - \sqrt{1-\beta^2})$
$= \frac{1}{2} \Delta t\, \beta^2$
$\simeq 0{,}47\,ns$

drift 0,125 ns OK

Ex 4:



$(\Delta t)_t = \gamma_t (\Delta t)_{in.}$

$(\Delta t)_a = \gamma_a (\Delta t)_{in}$

$= \gamma_a / \gamma_t \, (\Delta t)_t$

$(\Delta t)_t - (\Delta t)_a = (1 - \gamma_a/\gamma_t)(\Delta t)_t$

$\simeq \frac{1}{2} \cdot \frac{v_a^2 - v_t^2}{c^2} (\Delta t)_t$

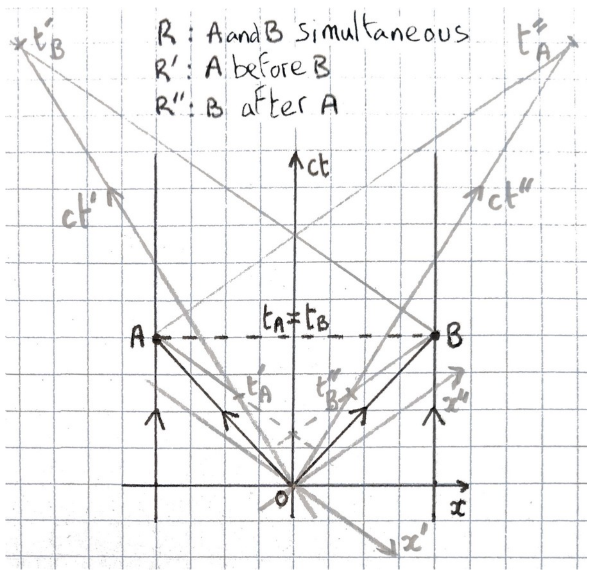
$v_t = 4.10^7/(24 \times 3600) \simeq 463\,m/s$

$v_a = \underbrace{v_{a/sol}}_{(-)} + v_t \simeq 741\,m/s$ $\Delta t \simeq 161\,ms$ Terrestrial Clock in advance

$\simeq 278\,m/s$ Toward the west: $\Delta t \simeq -86\,ns$ Terrestrial Clock Late

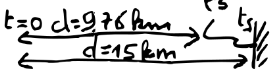
East	Ha/Hin	264ns	+161ns
	Ht/Hin	103ms	−86ns
West	Ha/Hin	16ms	

Diagrams for the flashlight in the middle of the wagon:



Exo 5: $P(t) = \lambda e^{-\lambda t}$; $p = P(T > t_{sol}) = \int_{t_s}^{+\infty} \lambda e^{-\lambda t}\, dt = e^{-d/H}$ $\lambda = 1/\gamma\tau_0$ $\tau = \gamma \tau_0$

$H = \gamma \tau \beta c \simeq 6575\,m$



15 km : $p \simeq 10{,}2\%$ 9760 m : $p \simeq 22.7\%$

(The memoryless law, on page 112 of the book by the same author : Probability, Stat. and Estimation)

For the train and tunnel paradox, see pages 75 and 76 of the book by the same author (fr):
Voyage pour Proxima, Comprendre Einstein plus vite que la lumière !